Minimax private density estimation with mixed anisotropy

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Abstract

We determine the minimax density-estimation rate under sequentially interactive local differential privacy and an anisotropic negative Sobolev loss, when the density and discriminator smoothness vectors are proportional. The result covers coordinates whose discriminator smoothness lies on opposite sides of the ambient dimension. The harmonic mean of discriminator smoothness determines the rate, with a single logarithmic factor at the critical value. A weighted randomized response for individual Fourier coefficients gives the upper bound. The noncritical lower bounds follow from Albert, Chevallier, Laurent and Sacko; at the critical value, an orthogonal information bound for the full private history yields the matching logarithmic lower bound.

1 The model and the rate

Local differential privacy requires each individual to release a randomized observation whose law changes by at most a prescribed likelihood ratio when that individual's data change. Density estimation under this constraint can be assessed through smooth test functions: the loss measures how well the estimated distribution reproduces their expectations. Different coordinates may require different degrees of smoothness.

Albert, Chevallier, Laurent and Sacko [1] study this problem with Fourier Sobolev density and discriminator classes. Their Theorem 10 gives an anisotropic lower bound when the two smoothness vectors are proportional. Their Theorem 11 gives upper bounds when all discriminator smoothness coordinates lie below the dimension, all lie above it, or all equal it. The paragraph following that theorem, on p. 15, leaves the mixed case open. Here we retain the sum of the anisotropic Fourier weights and sample coefficients with probabilities adapted to that sum. This supplies an upper bound in the mixed case and leads to a precise critical logarithm. The noncritical lower theorem is used with its original hypotheses and credit; the critical lower argument is given below.

Fix an integer $d \geq 2$, $R^2 > d$, $A > 0$, and vectors $\boldsymbol{\beta}, \boldsymbol{\delta} \in \mathbb{N}_{>0}^d$ such that $\beta_m/\delta_m = c > 0$ is independent of m. Assume that some $\delta_m < d$ and some $\delta_k > d$. Other coordinates may equal d. On $[0,1]$, use the real Fourier basis

$$\phi_1(t) = 1, \qquad \phi_{2j}(t) = \sqrt{2}\cos(2\pi jt), \qquad \phi_{2j+1}(t) = \sqrt{2}\sin(2\pi jt), \quad j \geq 1.$$

Its tensor products are $\phi_{\mathbf{j}}(x) = \prod_{m=1}^d \phi_{j_m}(x_m)$, with $\mathbf{j} \in \mathbb{N}_{>0}^d$. All integrals below use Lebesgue measure on $[0,1]^d$. Write $\theta_{\mathbf{j}}(h) = \int h\phi_{\mathbf{j}}$ and, for a positive smoothness vector $\mathbf{s}$, define

$$W^{\mathbf{s}}(r) = \left\{ h \in L^2([0,1]^d) : \sum_{\mathbf{j}} \left(\sum_{m=1}^d j_m^{2s_m} \right) |\theta_{\mathbf{j}}(h)|^2 \leq r^2 \right\}.$$

Let $\mathcal{F}_{\beta,R}$ be the probability densities in $W^{\beta}(R)$, and set

$$d_{\boldsymbol{\delta}}(h,f) = \sup_{g \in W^{\boldsymbol{\delta}}(1)} \left| \int (h-f)g \right|.$$

The data $X_1, \dots, X_n$ are independent with common density f. A sequentially interactive mechanism releases Z_i through a probability kernel $Q_i(\cdot \mid X_i, Z_{<i})$. Its α-privacy requirement is

$$Q_i(B \mid x, z_{<i}) \le e^{\alpha} Q_i(B \mid x', z_{<i}) \tag{1}$$

for every measurable output event B, every x, x', and every complete history $z_{<i}$. Denote the class of such mechanisms by $\mathcal{Q}_{\alpha}$. The risk of interest is

$$\mathcal{R}_{n,\alpha}(\boldsymbol{\beta}, \boldsymbol{\delta}) = \inf_{Q \in \mathcal{Q}_{\alpha}} \inf_{\widehat{f}} \sup_{f \in \mathcal{F}_{\boldsymbol{\beta},R}} \mathbb{E}_{f,Q} d_{\boldsymbol{\delta}}(\widehat{f}, f), \tag{2}$$

where $\widehat{f}$ is a measurable L^2-valued function of the released data. The loss in (2) is an expected norm, without squaring.

Theorem 1. *Under the fixed parameter assumptions above, put*

$$T = n\alpha^2, \qquad q = \sum_{m=1}^{d} \frac{1}{\delta_m}, \qquad c = \frac{\beta_m}{\delta_m}.$$

There are constants $C_-, C_+ > 0$, depending only on $d, R, A, \boldsymbol{\beta}, \boldsymbol{\delta}$, such that for every integer $n \ge 1$ and every $0 < \alpha \le A$ with $T \ge 1$,

$$C_- r(T) \le \mathcal{R}_{n,\alpha}(\boldsymbol{\beta}, \boldsymbol{\delta}) \le C_+ r(T), \qquad r(T) = \begin{cases} T^{-(c+1)/(2(c+q))}, & q > 1, \\ T^{-1/2} \log(eT), & q = 1, \\ T^{-1/2}, & q < 1. \end{cases} \tag{3}$$

Thus the three cases correspond to the harmonic mean $\bar{\delta} = d/q$ lying below, at, or above d. The power exponent can also be written $(\bar{\beta} + \bar{\delta})/(2\bar{\beta} + 2d)$, where $\bar{\beta}^{-1} = d^{-1} \sum_m \beta_m^{-1}$. The theorem concerns known, fixed smoothness parameters and pure local privacy. No adaptation to unknown smoothness is asserted.

2 Fourier weights and a private estimator

We suppress boldface on a Fourier multi-index j. Define

$$u_j = \sum_m j_m^{2\beta_m}, \qquad v_j = \sum_m j_m^{2\delta_m}.$$

The constant index is $\mathbf{1} = (1, \dots, 1)$. A density has $\theta_{\mathbf{1}} = 1$ and $u_{\mathbf{1}} = v_{\mathbf{1}} = d$. For every $h, f \in L^2$ the exact dual norm is

$$d_{\boldsymbol{\delta}}(h,f)^2 = \sum_j \frac{|\theta_j(h-f)|^2}{v_j}. \tag{4}$$

Weighted Cauchy–Schwarz gives one inequality. For a finite set E, the Fourier polynomial with coefficients proportional to $\theta_j(h-f)/v_j$, $j \in E$, attains the square root of the corresponding partial

sum after normalization in $W^{\boldsymbol{\delta}}(1)$. Increasing finite sets proves the reverse inequality. The full sum is finite because $v_j \geq d$. Also, comparison with the largest summand gives

$$d^{-c} v_j^c \leq u_j \leq d v_j^c. \tag{5}$$

The coefficient sampling probabilities will depend on the following sum. For $t > 0$, let

$$E_t = \{j \neq \mathbf{1} : v_j \leq t\}, \qquad N(t) = |E_t|, \qquad S(t) = \sum_{j \in E_t} v_j^{-1/2}.$$

Throughout the proof, comparison constants depend only on the fixed parameters, never on n, α, or the chosen mechanism.

Lemma 2. *For all sufficiently large t,*

$$N(t) \asymp t^{q/2}, \qquad S(t) \asymp \begin{cases} t^{(q-1)/2}, & q > 1, \\ \log t, & q = 1, \\ 1, & q < 1. \end{cases}$$

Proof. The inequalities $j_m \leq t^{1/(2\delta_m)}$ give the upper count. For the lower count, every point in the box

$$1 \leq j_m \leq \left\lfloor (t/d)^{1/(2\delta_m)} \right\rfloor, \qquad 1 \leq m \leq d,$$

has $v_j \leq t$. For large t, each floor is at least half its argument; removing the constant index leaves a count comparable to $t^{q/2}$.

Dyadic shells and the upper count bound the contribution at weight scale s by a constant times $s^{(q-1)/2}$. Summing these bounds gives the three upper estimates. To obtain lower estimates, let $a_0 t^{q/2} \leq N(t) \leq a_1 t^{q/2}$ for large t and choose a fixed $H > 1$ with $a_1 H^{-q/2} \leq a_0/2$. Then

$$N(t) - N(t/H) \geq (a_0/2) t^{q/2}$$

once both arguments are large. The shell $(t/H, t]$ therefore contributes at least $(a_0/2)t^{(q-1)/2}$ to $S(t)$. For $q > 1$ use the top shell. For $q = 1$ use the disjoint shells down to a fixed threshold; their number is comparable to $\log t$. For $q < 1$, one fixed nonconstant index gives the required positive lower bound. □

Choose a fixed t_0 large enough for Lemma 2 and for $E_{t_0} \neq \varnothing$. On E_t, $t \geq t_0$, set

$$p_j = \frac{v_j^{-1/2}}{S(t)}, \qquad B_0 = 2^{d/2}, \qquad \kappa = \frac{e^\alpha - 1}{e^\alpha + 1}. \tag{6}$$

An individual draws J with law (p_j), independently of X, and releases (J, Y), where $Y \in \{-1, 1\}$ has conditional law

$$\Pr(Y = y \mid X = x, J = j) = \frac{1}{2}\left(1 + \frac{y\kappa\phi_j(x)}{B_0}\right). \tag{7}$$

Use independent randomization for the n individuals and define

$$\widehat{\theta}_j = \frac{B_0}{n\kappa p_j} \sum_{i=1}^{n} \mathbf{1}_{\{J_i = j\}} Y_i, \qquad \widehat{f} = 1 + \sum_{j \in E_t} \widehat{\theta}_j \phi_j. \tag{8}$$

This estimator is a measurable finite Fourier polynomial.

Proposition 3. *The channel (7) belongs to $\mathcal{Q}_\alpha$, and*

$$\sup_{f\in\mathcal{F}_{\beta,R}} \mathbb{E}_{f,Q} d_{\boldsymbol{\delta}}(\widehat{f}, f) \le C\left(t^{-(c+1)/2} + \frac{S(t)}{\sqrt{T}}\right). \tag{9}$$

Proof. Since $|\phi_j(x)| \le B_0$, the ratio of any output-atom probabilities for two input points is at most $(1+\kappa)/(1-\kappa) = e^\alpha$. Summing over atoms proves privacy for all events. The channel is independent of the preceding history, so this inequality holds for every history in (1).

Each coefficient in (8) is unbiased. The squared single observation contribution has expectation $B_0^2/(\kappa^2 p_j)$, hence

$$\mathrm{Var}(\widehat{\theta}_j) \le \frac{B_0^2}{n\kappa^2 p_j}, \qquad \sum_{j\in E_t} \frac{\mathrm{Var}(\widehat{\theta}_j)}{v_j} \le \frac{B_0^2 S(t)^2}{n\kappa^2}.$$

The diagonal sum in (4) requires no independence between the estimated coordinates. The omitted-coefficient bias, by (5), is

$$\sum_{v_j>t} \frac{\theta_j(f)^2}{v_j} \le d^c t^{-(c+1)} \sum_j u_j \theta_j(f)^2 \le d^c R^2 t^{-(c+1)}.$$

Finally, $\kappa \ge \alpha/(e^A+1)$ because $e^\alpha - 1 \ge \alpha$ and $e^\alpha + 1 \le e^A + 1$. Apply Cauchy–Schwarz to the expectation of the norm to obtain (9). □

For $q > 1$, take $t = t_0 T^{1/(c+q)}$. For $q \le 1$, take $t = t_0 T^{1/(c+1)}$. Lemma 2 and Proposition 3 give the upper bound in (3), including $T = 1$.

3 An information bound with the full private history

The critical lower bound needs one information estimate shared by all the coefficients. Bounding each neighboring pair separately without this shared sum would miss the logarithm. We prove the estimate on joint transcript spaces, which also handles arbitrary measurable output spaces.

Lemma 4. *Let $(\phi_j)_{j\in E}$ be a finite mean-zero orthonormal family on the Lebesgue probability space $[0,1]^d$. Suppose $a_j > 0$ and every density*

$$f_\varepsilon = 1 + \sum_{j\in E} \varepsilon_j a_j \phi_j, \qquad \varepsilon \in \{-1,1\}^E,$$

lies between $1/2$ and $3/2$ almost everywhere. Fix any $Q \in \mathcal{Q}_\alpha$, and let M_ε^n be its full transcript law under f_ε. If $\varepsilon^{(j)}$ denotes the vector with sign j reversed, there are nonnegative numbers $\bar{I}_j$ such that

$$\sum_{j\in E} \bar{I}_j \le n(e^\alpha - 1)^2, \qquad \tau_j := 2^{-|E|} \sum_\varepsilon \mathrm{TV}(M_\varepsilon^n, M_{\varepsilon^{(j)}}^n) \le 2a_j\sqrt{\bar{I}_j}. \tag{10}$$

Here $\mathrm{TV}(P,Q) = \sup_B |P(B) - Q(B)|$.

Proof. Write M_0^i for the law of the first i releases under the uniform density, and let $h = z_{<i}$ denote the full history. Its uniform-input next-step kernel is $K_{0,i}(\cdot \mid h) = \int Q_i(\cdot \mid x, h)\, dx$. Integrating (1) over a uniform second input gives, for every x, h,

$$e^{-\alpha} K_{0,i}(\cdot \mid h) \le Q_i(\cdot \mid x, h) \le e^\alpha K_{0,i}(\cdot \mid h). \tag{11}$$

Introduce two finite measures on the joint space of (x, h, z):

$$\Gamma_i(dx, dh, dz) = dx\, M_0^{i-1}(dh) Q_i(dz \mid x, h), \qquad \Lambda_i(dx, dh, dz) = dx\, M_0^i(dh, dz).$$

Applying (11) to measurable sections and integrating gives $e^{-\alpha}\Lambda_i \le \Gamma_i \le e^{\alpha}\Lambda_i$. The Radon–Nikodym theorem and Fubini's theorem for finite measures [3, 232F, 252C, 252H] give a jointly measurable density $r_i = d\Gamma_i/d\Lambda_i$ satisfying

$$e^{-\alpha} \le r_i \le e^{\alpha}, \qquad \int r_i(x, h, z)\, dx = 1$$

for M_0^i-almost every (h, z). The last equality follows because the projection of Γ_i onto (h, z) is M_0^i. Set

$$s_{ij}(h, z) = \int \phi_j(x) r_i(x, h, z)\, dx, \qquad m_{i,\varepsilon}(h, z) = 1 + \sum_j \varepsilon_j a_j s_{ij}(h, z).$$

Then $m_{i,\varepsilon} = \int f_\varepsilon r_i\, dx \in [1/2, 3/2]$. Expanding the integral of the nonnegative square $(r_i - 1 - \sum_j s_{ij}\phi_j)^2$, using the mean-zero orthonormal family, yields the pointwise joint bound

$$\sum_j s_{ij}(h, z)^2 \le \int (r_i - 1)^2 dx \le (e^{\alpha} - 1)^2. \tag{12}$$

For each bounded measurable function F of the preceding history,

$$\int F(h) s_{ij}(h, z)\, dM_0^i(h, z) = 0. \tag{13}$$

Indeed, use Γ_i for this integral and integrate z first; the remaining integral of ϕ_j over x vanishes. The identity extends to integrable history weights by truncation, since s_{ij} is bounded.

Let $L_\varepsilon^i = dM_\varepsilon^i/dM_0^i$. The fresh X_i is independent of the preceding released history under each fixed density. Induction therefore gives

$$L_\varepsilon^i(h, z) = L_\varepsilon^{i-1}(h) m_{i,\varepsilon}(h, z), \qquad L_\varepsilon^0 = 1. \tag{14}$$

To check the identity, test its right-hand side against a bounded measurable function of (h, z), replace $m_{i,\varepsilon}$ by $\int f_\varepsilon r_i dx$, and use Γ_i and the preceding history density. This gives exactly the next-step law under f_ε. All L_ε^i are positive. This argument does not require measurable versions of conditional Radon–Nikodym kernels on the output space.

For neighboring signs, $m_{i,\varepsilon} - m_{i,\varepsilon^{(j)}} = 2\varepsilon_j a_j s_{ij}$. By (14), the logarithm of the full likelihood ratio is the sum of the corresponding logarithms of the m's. For positive a, b,

$$a \log(a/b) \le \frac{(a-b)^2}{b} + (a - b).$$

The last term has zero integral against the preceding history density by (13). The denominator $m_{i,\varepsilon^{(j)}}$ is at least $1/2$, so

$$\mathrm{KL}(M_\varepsilon^n, M_{\varepsilon^{(j)}}^n) \le 8 a_j^2 I_j(\varepsilon), \qquad I_j(\varepsilon) = \sum_{i=1}^n \int L_\varepsilon^{i-1}(h) s_{ij}(h, z)^2\, dM_0^i(h, z). \tag{15}$$

All logarithms are bounded at each step, and all these integrals are finite. Since $\int L_\varepsilon^{i-1}(h) dM_0^i = 1$, summing (12) gives

$$\sum_j I_j(\varepsilon) \le n(e^{\alpha} - 1)^2.$$

Define $\bar{I}_j$ as the uniform average of $I_j(\varepsilon)$.

In our normalization, $\mathrm{TV}(P,Q)^2 \le \mathrm{KL}(P,Q)/2$. Here this inequality can be seen directly by taking $B = \{dP/dQ \ge 1\}$ and writing $p = P(B)$, $q_0 = Q(B)$. The convex tangent inequality for $t \log t$ on B and its complement gives

$$\mathrm{KL}(P,Q) \ge p \log \frac{p}{q_0} + (1-p) \log \frac{1-p}{1-q_0} \ge 2(p-q_0)^2.$$

For the second inequality the expression as a function of p has value and first derivative zero at q_0, and second derivative $1/[p(1-p)] \ge 4$. Zero-mass endpoints are handled by continuity or infinite relative entropy. Also $p - q_0 = \mathrm{TV}(P,Q)$. Apply this inequality to (15) and use Cauchy–Schwarz when averaging over signs. This proves (10). □

The joint measures in this proof keep every preceding release in the history. The private releases need not be independent. The argument refines the use of privacy contraction in local-private minimax bounds [2], by retaining a sum over orthogonal coefficients at each step.

4 The critical lower bound and completion of the proof

At $q = 1$, Lemma 2 supplies a logarithm in the weighted spectral sum. We put a finite family of densities on those coefficients, with amplitudes that make the testing bound spend the same shared information sum. The cutoff can grow slowly: the lower bound needs a number of scales proportional to $\log T$, rather than the largest admissible cutoff.

Assume $q = 1$ and set

$$\rho = \frac{1}{4(c+1)}, \qquad t = T^{\rho}, \qquad E = E_t, \qquad S = S(t).$$

For T sufficiently large, define positive amplitudes by

$$a_j^2 = \frac{\eta^2 S \sqrt{v_j}}{T}, \qquad j \in E, \tag{16}$$

where $\eta > 0$ is fixed below. Lemma 2 gives $N(t) \lesssim t^{1/2}$ and $S \asymp \log t \asymp \log T$. Consequently,

$$B_0 \sum_{j \in E} a_j \lesssim \eta \sqrt{S/T}\, t^{3/4} \longrightarrow 0, \tag{17}$$

$$\sum_{j \in E} u_j a_j^2 \lesssim \eta^2 (S/T) t^{c+1} = \eta^2 S T^{-3/4} \longrightarrow 0. \tag{18}$$

The first convergence uses $\rho < 1/4$, and the second uses (5). Since $R^2 > d$, there is a fixed T_* such that for every $T \ge T_*$ and every sign vector the Fourier polynomial

$$f_\varepsilon = 1 + \sum_{j \in E} \varepsilon_j a_j \phi_j$$

lies between $1/2$ and $3/2$ and has Sobolev weight $d + \sum_j u_j a_j^2 \le R^2$. Each nonconstant basis function has mean zero, so all these polynomials are probability densities in the original class. The positivity is uniform over every sign vector.

Fix any mechanism $Q \in \mathcal{Q}_\alpha$ and any measurable L^2-valued estimator. Decode its jth Fourier coefficient as $+1$ when it is nonnegative and as -1 otherwise. This is measurable, and a wrong decoded sign causes coefficient error at least a_j. Write

$$w_j = v_j^{-1}, \qquad b_j = w_j a_j^2, \qquad V = \sum_{j \in E} b_j.$$

By (4), pointwise,

$$d_{\boldsymbol{\delta}}(\widehat{f}, f_\varepsilon) \geq \sqrt{\sum_j b_j \mathbf{1}_{\{\widehat{\varepsilon}_j \neq \varepsilon_j\}}} \geq \frac{1}{\sqrt{V}} \sum_j b_j \mathbf{1}_{\{\widehat{\varepsilon}_j \neq \varepsilon_j\}}. \tag{19}$$

The second inequality uses $\sqrt{x} \geq x/\sqrt{V}$ for $0 \leq x \leq V$. For a neighboring pair of transcript laws, the sum of the two decision error probabilities is at least $1 - \mathrm{TV}$. Thus the sign-average risk is at least

$$\frac{V - \sum_j b_j \tau_j}{2\sqrt{V}}, \tag{20}$$

where τ_j is from Lemma 4. This is a lower bound for the expected norm itself.

For the amplitudes (16), direct summation gives

$$V = \frac{\eta^2 S^2}{T}, \qquad \sum_j b_j^2 a_j^2 = \frac{\eta^6 S^4}{T^3}.$$

Moreover $e^\alpha - 1 \leq e^A \alpha$. Weighted Cauchy–Schwarz and (10) therefore yield

$$\sum_j b_j \tau_j \leq 2 \left(\sum_j b_j^2 a_j^2 \right)^{1/2} \left(\sum_j \bar{I}_j \right)^{1/2} \leq \frac{2 e^A \eta^3 S^2}{T}.$$

Choose $0 < \eta \leq 1/(4e^A)$. The last quantity is at most $V/2$. Equation (20) now gives a lower bound $\eta S/(4\sqrt{T}) \gtrsim \log T/\sqrt{T}$ for every mechanism and estimator, uniformly for $T \geq T_*$. The finite density family and all constants were chosen independently of the mechanism.

Proof of Theorem 1. The upper bound was established after Proposition 3. For the remaining lower estimates, Theorem 10 of [1, p. 14] states, under the same positive integer smoothness, proportionality, radius and privacy assumptions,

$$\mathcal{R}_{n,\alpha}(\boldsymbol{\beta}, \boldsymbol{\delta}) \geq C \max \left\{ T^{-(\bar{\beta}+\bar{\delta})/(2\bar{\beta}+2d)}, T^{-1/2} \right\}.$$

Its mechanism class uses the full-history privacy condition (1); its minimax loss is the expected unsquared adversarial distance. The absolute value in our definition gives the same distance, since the discriminator ball is symmetric. The real Fourier indexing and sum weights are also the same. As $\bar{\delta} = d/q$ and $\bar{\beta} = cd/q$, the first exponent is $(c+1)/(2(c+q))$. This proves the required lower bound for $q > 1$ and $q < 1$. When $q = 1$, the preceding construction proves the logarithmic lower bound for $T \geq T_*$. For $1 \leq T < T_*$, the cited bound is at least $CT^{-1/2}$, and $\log(eT) \leq \log(eT_*)$; decreasing the fixed lower constant covers this interval as well. Both lower arguments hold for every Q and every measurable L^2-valued estimator before taking the infima in (2). This proves (3) throughout the stated parameter range. □

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